EDS 223: Homework Assignment 2

title

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```
# Loading necessary packages
library(tidyverse)
library(sf)
library(here)
library(tmap)
library(spData)

# Read in geodatabase of EJScreen data at the Census Block Group level
ejscreen <- sf::st_read(here::here("data",
"ejscreen","EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"))

# Filter to California
ca <- ejscreen %>%
  dplyr::filter(ST_ABBREV == "CA")

# Filter to LA county
los_angeles <- ca %>%
  dplyr::filter(CNTY_NAME %in% c("Los Angeles County"))

# Find the average values for all variables within counties
ca_counties <- aggregate(ca, by = list(ca$CNTY_NAME), FUN = mean)


# Read in HOLC Redlining
mapping_ineq <- sf::st_read(here::here("data",
"mapping-inequality","mapping-inequality-los-angeles.json"))


# Read in bike and pedestrian crash reports
```

```
crashes <- sf::st_read(here::here("data",
"SWITRS_LA","crashes.csv"))
```

```
# Make sure to check that these datasets have the same coordinate reference systems!
# If not, transform them to match.

st_crs(los_angeles) == st_crs(mapping_ineq) # returns FALSE
```

```
[1] FALSE
```

```
mapping_ineq_transform <- st_transform(mapping_ineq, crs = st_crs(los_angeles))
```

```
# Checking if CRSs match
if(st_crs(los_angeles) == st_crs(mapping_ineq_transform)){
  print("It's a match!")
}else {
  print("Still not a match.")
}
```

```
[1] "It's a match!"
```

Part 1: Legacy of redlining in current environmental (in)justice

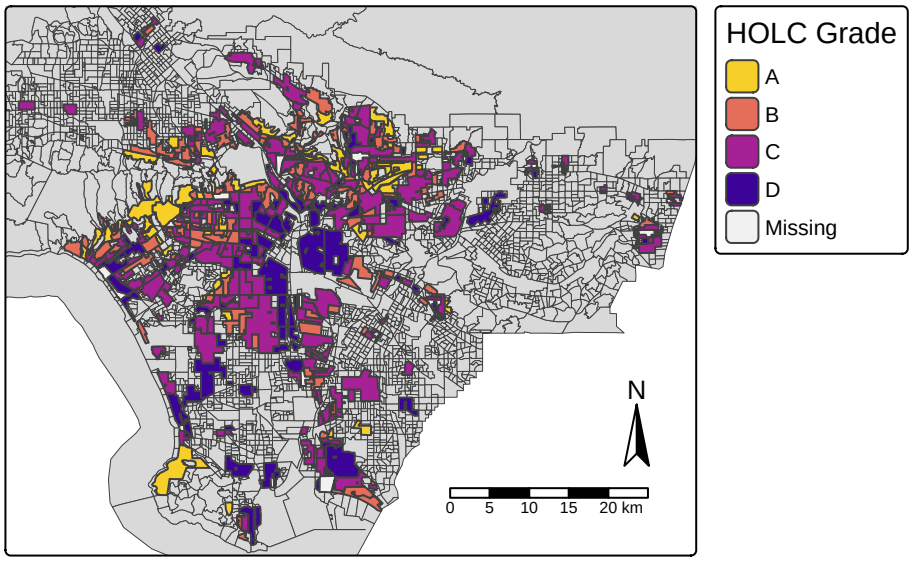
1. Create a map of historical redlining neighborhoods, including:

- neighborhoods colored by HOLC grade
- an appropriate base map
- a clear title and legend

```
tmap_mode('plot')

# Map of historical redlining neighborhoods
tm_shape(los_angeles) +
    tm_polygons(lwd = 0.15) +
tm_shape(mapping_ineq_transform, is.main = TRUE) +
    tm_polygons(fill = 'grade',
                fill.scale = tm_scale(values = "-plasma"),
                fill.legend = tm_legend(title = 'HOLC Grade')) +
    tm_title(text = 'Redlined Neighborhoods in Los Angeles County with HOLC Grade ') +
tm_compass() +
tm_scalebar(position = c('bottom', 'right'))
```

Redlined Neighborhoods in Los Angeles County with HOLC Grade



2. Create a table summarizing:

- the percentage of census block groups that fall within each HOLC grade
- Also include the percent of census block groups that don't fall within a HOLC grade in your table
- **Hint:** The HOLC data contains the grades and the EJScreen data contains the census blocks, so you will need to combine the data spatially before doing summary statistics. Once you combine and no longer need the geometries, you can use `st_drop_geometry()`.

```
df_joined_dropped <- st_join(los_angeles, mapping_ineq_transform) |>  # left joining data
    st_drop_geometry() |>  # dropping geometries
    count(grade) |> # finding total census blocks per HOLC grade
    mutate(percent_block_per_grade = round((n / sum(n) * 100), 2)) # adding a percentage colu
df_joined_dropped
```

```
  grade    n percent_block_per_grade
1     A  449                    5.00
2     B 1239                   13.79
3     C 3058                   34.02
4     D 1346                   14.98
5  <NA> 2896                   32.22
```

3. Create at least two visualizations summarizing current conditions (from the EJScreen data) within HOLC grades using the mean of the following variables (you may combine variables in a single visualization or look at a single variable per plot):

- % low income
- percentile for Particulate Matter 2.5
- percentile for low life expectancy
- Use `ggplot` for your visualizations! You will first need to calculate mean of each variable grouped by HOLC grade.

4. Write a brief paragraph reflecting on these results

- Interpret the patterns you observe in your results
- Discuss potential relationships between historical redlining grades and current environmental/socioeconomic conditions